

CMP5367 Artificial Intelligence and Machine Learning A S1 2025/6

Lecture 2:

Data Exploration and Preprocessing

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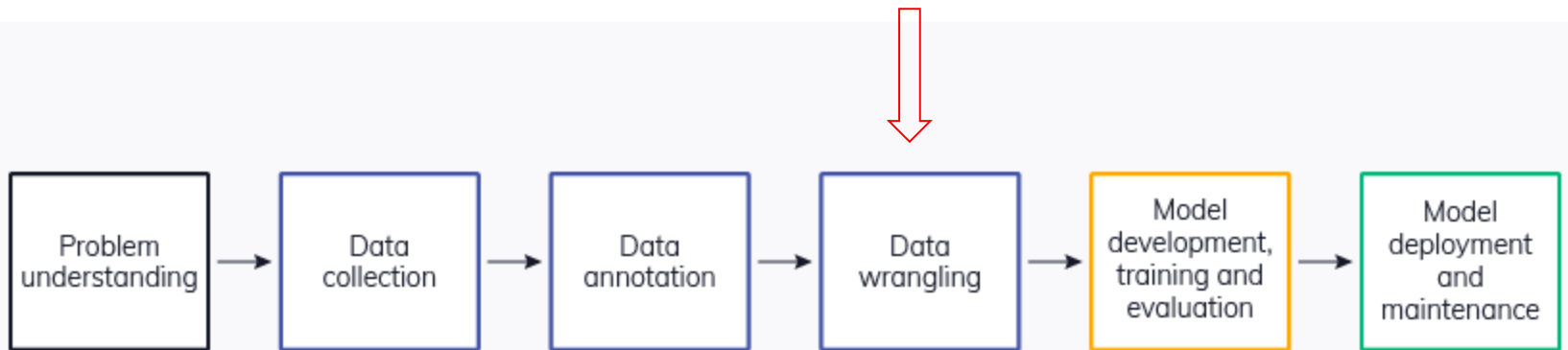
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September 29th , 2025

Topics

- Common dataset challenges.
- Data cleaning.
- Data transformation.
- Data splitting for ML.
- Bias and Variance.

Where are we in the Machine Learning Lifecycle



Common dataset challenges

Poor data quality is enemy number one to machine learning. If you want machine learning to work for you, you need a sharp focus on data quality.



➤ **Accessibility:**

Privacy, legal issues, health records, commercial reputation, competitiveness, etc.

➤ **Inaccuracy:**

Human errors, data drift, and data decay, poor labelling, etc.

➤ **Inconsistency:**

Multiple data sources, mismatches in the same information across sources, discrepancies in formats, or units, or sometimes spellings.

Common dataset challenges



Clifford   @holte · Sep 24

Oh dear..., I fear the wurst....

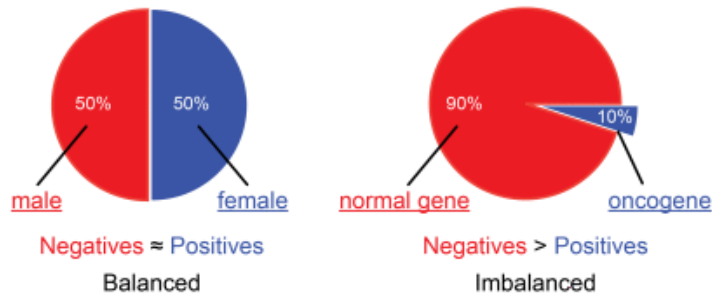
 24

 28

 1.8K

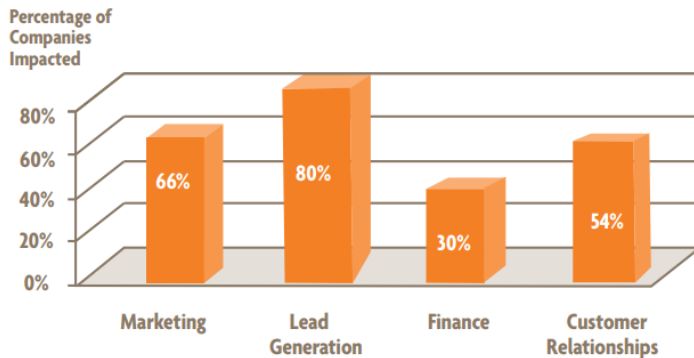
Needs cleaning

Example of balanced and imbalanced data



Unbalanced

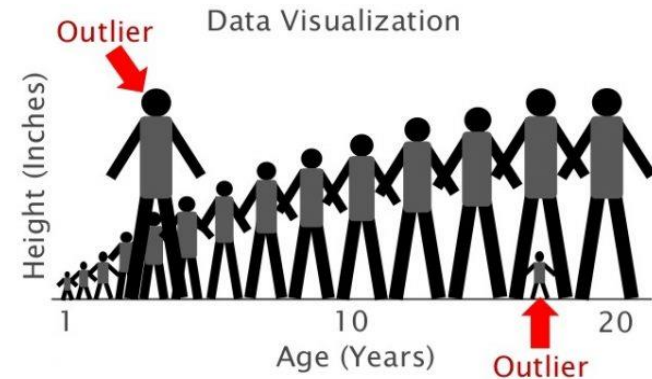
Top areas impacted by inaccurate data



Inaccurate data

Bronx Rolling Sales File. All Sales From April 2020 - March 2021		Unnamed: 1	Unnamed: 2	Unnamed: 3
0	For sales prior to the Final Roll, Neighborhood...	NaN	NaN	NaN
1	Building Class Category is based on Building C...	NaN	NaN	NaN
2	Note: Condominium and cooperative sales are on...	NaN	NaN	NaN
3	BOROUGH NEIGHBORHOOD BUILDING CLASS CATEGORY TAX CLASS AT PRESENT			
4	2 BATHGATE 01 ONE FAMILY DWELLINGS 1			

Missing values



Outliers

	A	B
1	Name	Is Duplicate
2	Sigmund Augdal Helberg	TRUE
3	Marc Ariberti	FALSE
4	Stephan Assmus	TRUE
5	Olivier Aubert	FALSE
6	Sigmund Augdal Helberg	TRUE
7	Pierre Baillet	FALSE
8	Gildas Bazin	FALSE
9	Rocky Bernstein	TRUE
10	Johan Bilien	FALSE

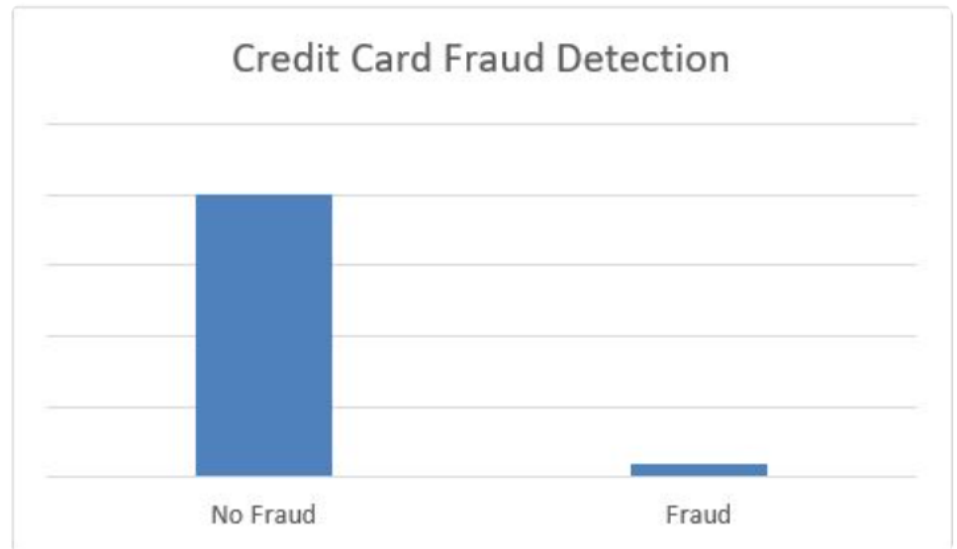
Duplicates

Data Processing

Unbalanced Data: (majority and minority class)

What would problems may occur in training on this data?

- a) Skewed Class Distribution
- b) Poor Generalization
- c) Costly Errors
- d) Evaluation Metrics Misleading





What would be a bias in a dataset? How can this affect training an ML model?



COOKING

ROLE	VALUE
AGENT	▶ WOMAN
FOOD	▶ PASTA
HEAT	▶ STOVE
TOOL	▶ SPATULA
PLACE	▶ KITCHEN



COOKING

ROLE	VALUE
AGENT	▶ WOMAN
FOOD	▶ FRUIT
HEAT	▶ —
TOOL	▶ KNIFE
PLACE	▶ KITCHEN



COOKING

ROLE	VALUE
AGENT	▶ WOMAN
FOOD	▶ MEAT
HEAT	▶ GRILL
TOOL	▶ TONGS
PLACE	▶ OUTSIDE



COOKING

ROLE	VALUE
AGENT	▶ WOMAN
FOOD	▶ VEGETABLES
HEAT	▶ STOVE
TOOL	▶ TONGS
PLACE	▶ KITCHEN



COOKING

ROLE	VALUE
AGENT	▶ MAN
FOOD	▶ —
HEAT	▶ STOVE
TOOL	▶ SPATULA
PLACE	▶ KITCHEN

In this example of gender bias, adapted from a report published by researchers from the University of Virginia and the University of Washington, a visual semantic role labeling system has learned to identify a person cooking as female, even when the image is male.



Solutions of Sparse/Missing data

1. **Convert the feature from sparse to dense** (data imputation)
 1. Use the mean or median value of a particular feature.
 2. Use the value from the previous or next data point
 3. Resampling. Generate many random samples from the data.

BUT

There is no fit-all solution

In certain contexts, it does not make sense at all to use the mean. For example:

- a. All your values are integers, and the mean is a fraction.
- b. Using only the mean can add too much bias toward the centre.

Don't forget to evaluate the effectiveness of your chosen method.

Common Data Cleaning Techniques

Handling Missing Values:

Imputation:

- ✓ *Replace missing values with substituted values:*
- ✓ *Mode*
- ✓ *Mean*
- ✓ *Most frequent category*
- ✓ *More sophisticated (regression imputation, etc.)*
- ✓ *Or just.....*
- ✓ *Deletion*

Descriptive Statistics:

Average / Mean

“Arithmetic Mean”

Average = sum / total number samples

$$\text{Average}(1,2,3) = (1+2+3) / 3 = 6/3 = 2$$

Median

Midpoint of a dataset

Value	11	22	33	44	55
Sample	1	2	3	4	5

In the example above, Median = 33

Question: What if the
values are even?

Mode

- **Mode** is simply the most common value

- Attribute data:

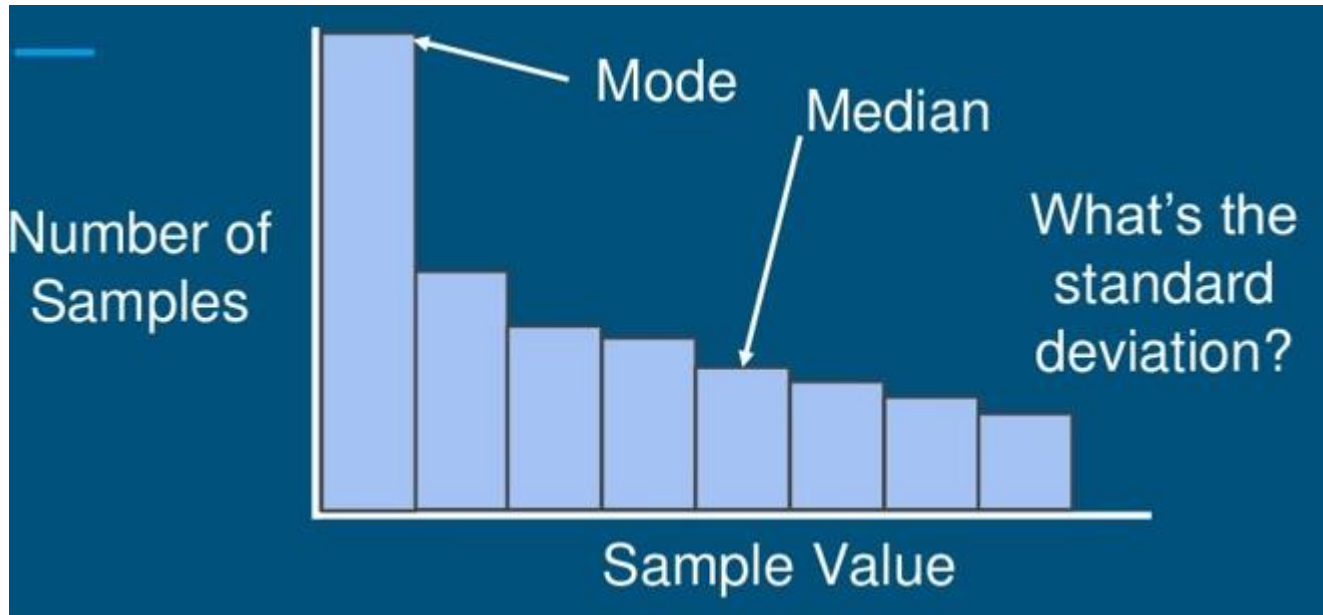
1,1,2,2,2,3,3,3,3,4,5,5,6,6,7,8,10,13

- Mode of the above data: 3

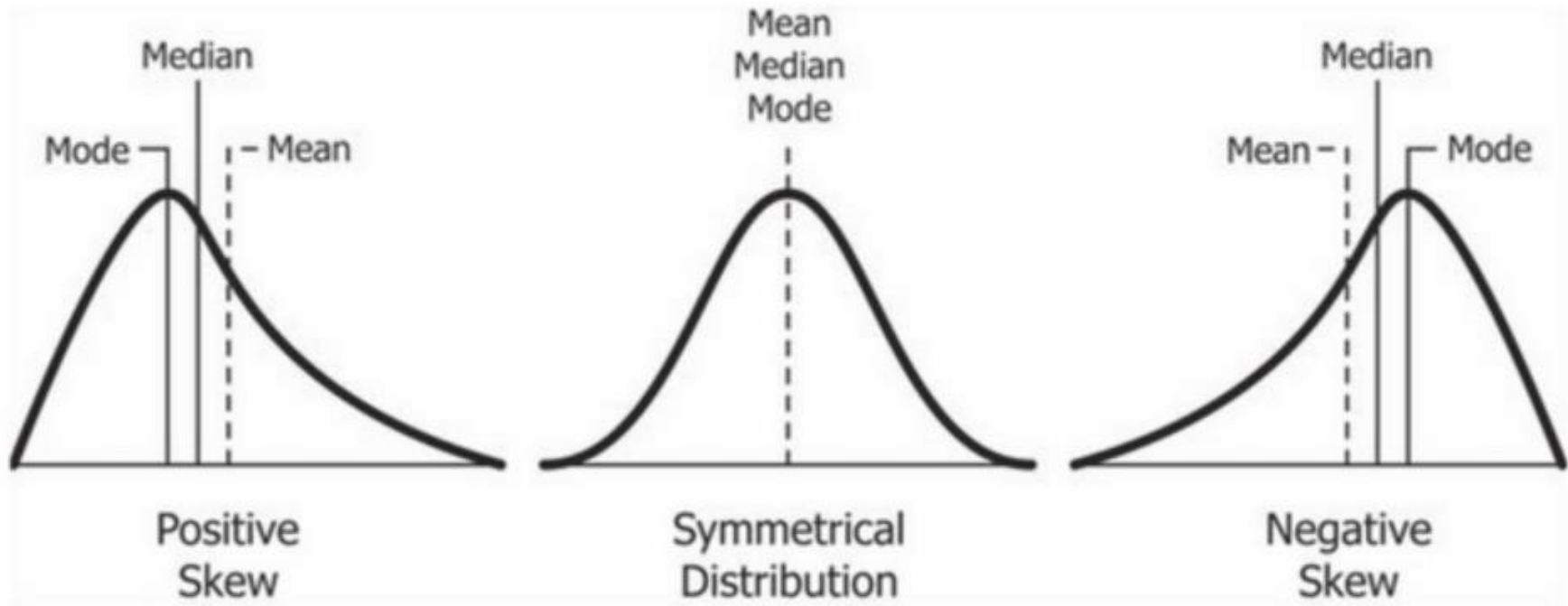
Quiz



Scaling is important for Non-Normal Distribution



Skewness and filling in missing data



Skewness Type	Description	Best Imputation Strategy
Near 0	Symmetric (Gaussian-like)	Mean
Positive (>1)	Long tail to the right (e.g. prices)	Median or log transform + median
Negative (←1)	Long tail to the left	Median or mode

Data Scaling

There are several ways to achieve data scaling:

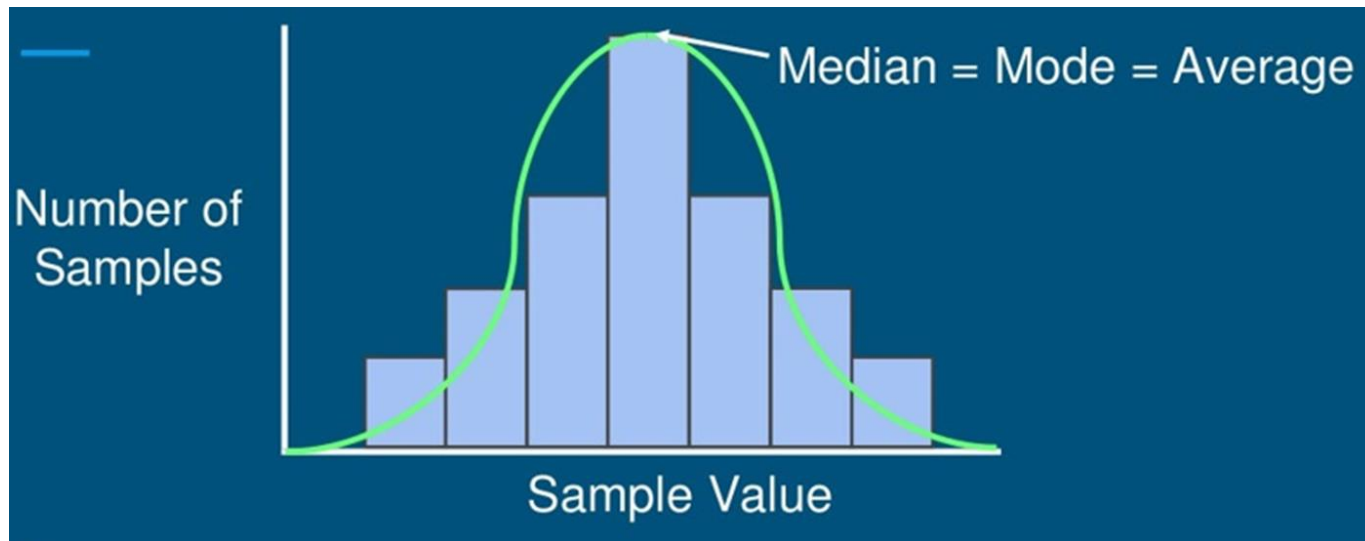
Option 1:

Min-Max normalization scales the data within a fixed range (usually 0 to 1)

Option 2:

Standardization scales data with a mean of 0 and a standard deviation of 1.

You try to have the data in normally distributed:



USE CASE

If I have this range: **Sale Price Range: £0.00 to £27,018,000.00**

➤ **Step 1: Investigate Distribution and Outliers**

Look for **long tails**, **spikes at zero**.

Confirm if zeros are valid or placeholders for missing data (do we really have houses with 0 price?).

➤ **Step 2: Handle Invalid Zeros (if needed)**

Replace 0s with NaN if they represent missing values

➤ **Step 3: Impute**

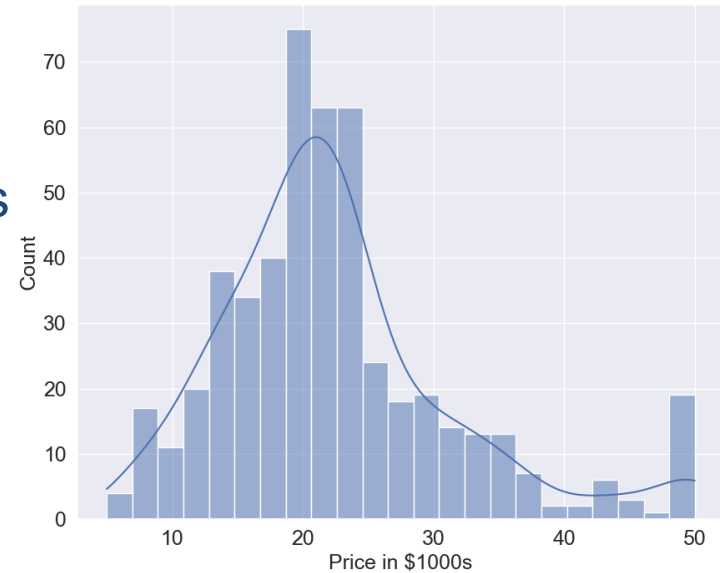
Use **median** to avoid distortion from outliers

➤ **Step 4: Apply Log Transformation (to reduce skew)**

`log1p()` handles zeros safely.

This transformation compresses large values and spreads out small ones.

➤ **Step 5: Scale for Modeling**



Standard Deviation (σ)

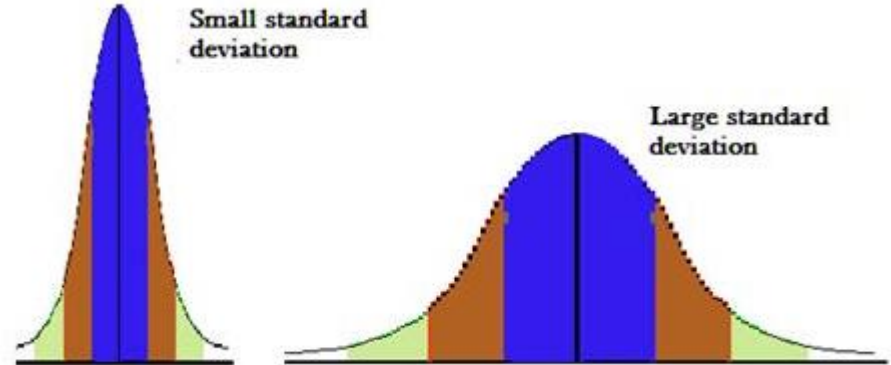
$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

where:

x_i = Value of the i^{th} point in the data set

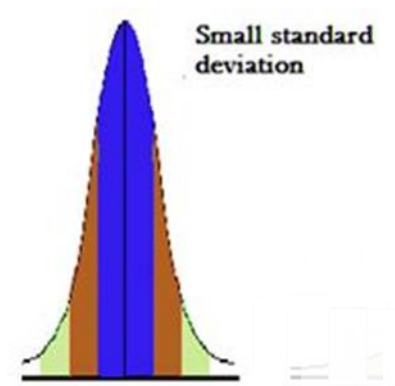
$\bar{x}$ = The mean value of the data set

n = The number of data points in the data set



1. Calculate the mean of all data points. The mean is calculated by adding all the data points and dividing them by the number of data points.
2. Calculate the variance for each data point. The variance for each data point is calculated by subtracting the mean from the value of the data point.
3. Square the variance of each data point (from Step 2).
4. Sum of squared variance values (from Step 3).
5. Divide the sum of squared variance values (from Step 4) by the number of data points in the data set less 1.
6. Take the square root of the quotient (from Step 5).

What Does Standard Deviation Mean in a Dataset?



When a Small Standard Deviation Is Good?

Use case:

When you want the model to learn subtle patterns in a tightly controlled environment.

- **Example:** Predicting house prices in a **small town** where most homes are similar in size, age, and location.
- A small standard deviation in features like **square footage** or **number of bedrooms** helps the model converge (**reach optimum performance**) quickly and generalize well within that domain.
- **Benefit:** Easier to train, less risk of overfitting to outliers or noise.

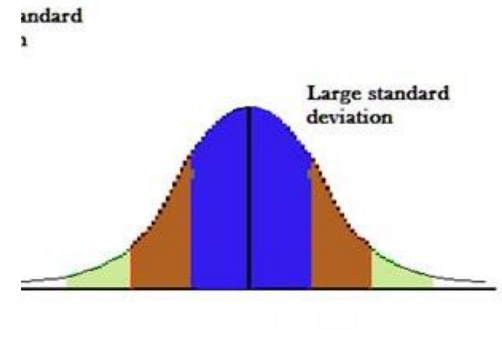
What Does Standard Deviation Mean in a Dataset?

When a Large Standard Deviation Is Good?

Use case:

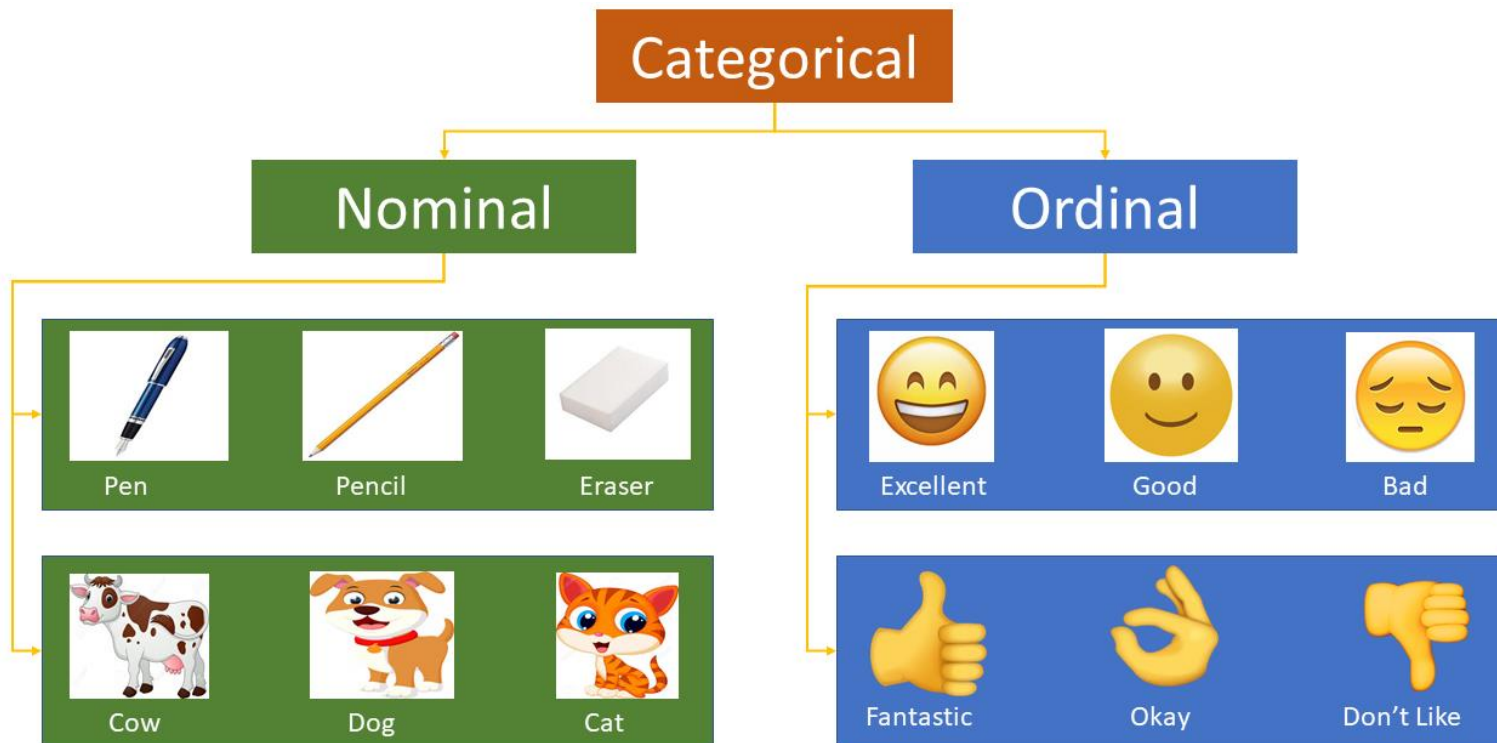
When you want the model to generalize across diverse conditions or populations.

- **Example:** Sentiment analysis on **social media posts from users across different countries, age groups, and writing styles..**
- A large standard deviation in vocabulary usage, sentence length, or sentiment scores reflects real-world diversity.
- **Benefit:** Helps the model learn robust features that generalize across varied inputs.



Converting Categorical Data into Numerical Form

Categorical variables (qualitative or discrete variables):



Challenges Faced with Categorical Variables in Machine Learning

- ❑ **Misinterpretation of Category Encoding:**

1 for 'Category A', 2 for 'Category B', is category B twice as significant as category A)

- ❑ **High Cardinality:**

'City' variable in a global dataset (large, sparse matrix that is computationally expensive)

Techniques for encoding categorical data

- **Label Encoding (not the best for nominal variables why?)**

```
(['Red', 'Blue', 'Green', 'Red', 'Green'])    [1 0 2 1 2]
```

- **One-Hot Encoding (good for nominal variables why?)**

	Is_Red	Is_Blue	Is_Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1
Blue	0	1	0
Red	1	0	0

- **Binary Encoding (good for high cardinality cases)**

```
Red:    1 -> 01
Blue:   2 -> 10
Green:  3 -> 11
Yellow: 4 -> 100
```


Data cleaning vs. data transformation (Feature engineering)

Encoding:

Name	Gender	isFemale?
James	M	0
Juliet	F	1
Jane	F	1
Brown	M	0

Data binning (Data Discretization)

Name	Gender	Age	Age Group
James	M	15	Adolescent
Juliet	F	17	Adolescent
Jane	F	25	Adult
Brown	M	28	Adult

Bins
<=18: Adolescent
>18: Adult

Scaling:

Country	Age	Salary	Scaled Salary
France	44	72000	0.6857
Spain	27	48000	0.0000
Germany	30	54000	0.1714
Spain	38	61000	0.3714
France	35	58000	0.2857
France	48	79000	0.8857
Germany	50	83000	1.0000
France	37	67000	0.5429

Quiz

- How many preprocessing techniques can you use for this dataset?

Employee Id	Name	Age	Salary	Gender	Status
101	Richard Harris	20	19000	M	Full-time
102	David Harman	60	45000	M	Full-time
103	Tom Patrick	22	54000	M	Full-time
104	Lisa Hans	50	70000	F	Full-time
105	Kate Harrison	55	83000	F	Full-time
106	George Samuel	28	15000	M	Part-time
112	Ron Tomlinson			M	Part-time
101	Richard Harris	20	19000	M	Part-time
104	lisa hans	50	70000	F	Full-time
111	Hu Lee	30	22000	M	Full-time
112	Juliet Brown	35	35000	F	Full-time

Summary

- Data Challenges
- Data Cleaning
- Data Transformation
- Scaling
- Categorical Data Imputation